A PROJECT REPORT
ON

Pick and Place Robotic Arm

BACHELOR OF TECHNOLOGY UNDER VIT CHENNAI UNIVERSITY

VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

School of Electronics Engineering, VIT Chennai

MICROPROCESSORS AND MICROCONTROLLERS (BECE204L)
SLOT : B1+TB1

Submitted by:-

23BEC1136-Harini Kamatchi V
23BLC1043-Mridula B
23BLC1183-Lakshitha V
23BLC1287-Kiran Pradha M S
23BEC1198-Alivia Das

Submitted to:-

Dr. Chanthini Baskar

B.Tech (ECE,ECM)
IInd - year

Abstract

The Pick and Place Robotic Arm project is all about creating an automated system that can pick up items from one spot and place them accurately somewhere else. This robotic arm is driven by a 8051 microcontroller and uses servo motors to move precisely in different directions. It features a gripper mechanism that securely holds onto objects, while sensors such as infrared or ultrasonic modules help with detecting and positioning those items. The system is designed to handle repetitive tasks efficiently, which cuts down on manual labour and boosts productivity, particularly in industrial and lab environments. Its uses range from simple automation in packaging to educational demos and prototyping. This project highlights the fundamental concepts of robotics, electronics, and automation, laying the groundwork for future upgrades like image processing, AI-driven object recognition, and IoT integration for remote operation. It provides a cost-effective, scalable solution for smart automation.

Introduction

Automation has become an essential part of today’s industries, helping to boost productivity, improve accuracy, and cut down on manual labor. Among the various automated systems, robotic arms are particularly important for handling repetitive tasks like picking and placing items. This project is all about designing and developing a basic Pick and Place Robotic Arm that can be controlled wirelessly through Bluetooth.

The robotic arm operates using servo motors, which provide smooth and precise movements, and is managed by an 8051 microcontroller. A Bluetooth module allows it to receive commands from a mobile device, giving users the ability to control the arm from a distance. It also features a simple gripper mechanism for picking up and placing lightweight objects.

This project showcases fundamental concepts in robotics, wireless communication, and embedded systems. It offers a budget-friendly and practical solution for basic automation tasks and lays the groundwork for more advanced innovations in the robotics field.

Components

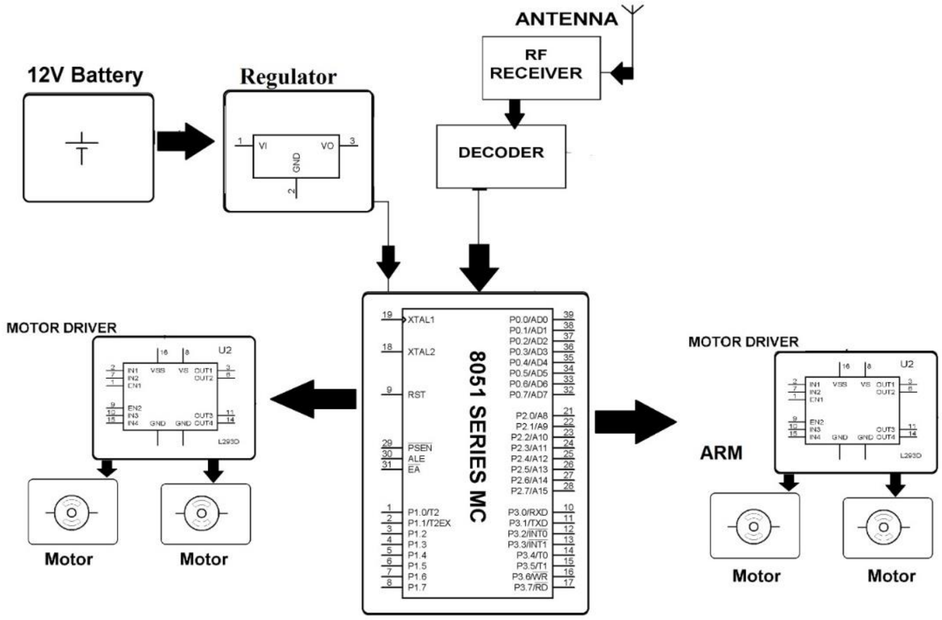
Hardware:

- 8051 Microcontroller
- Bluetooth Module
- Jumper wires
- Servo motors
- Motor driver
- 12V battery supply
- ISP Programmer
- Robotic arm

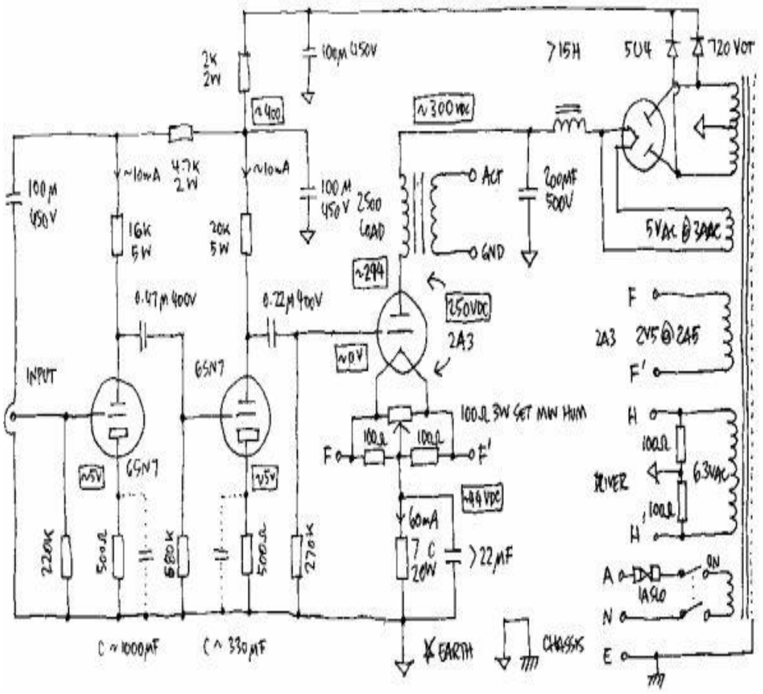
Software:

- Keil µvision
- Progisp
- Zadig

Methodology



- A 12V battery powers the entire system, which is then stepped down to 5V through a voltage regulator to safely supply the microcontroller and other components.
- An RF receiver, equipped with an antenna, picks up wireless signals from a remote RF transmitter.
- The decoder interprets these RF signals and sends the command over to the 8051 microcontroller.
- The 8051 microcontroller takes care of processing the command and dispatches the right control signals to the motor drivers.
- To manage four DC motors, we use two L293D motor driver ICs—one for the base and another for the robotic arm.
- Each motor driver gets low-power control signals from the microcontroller and boosts them to drive the motors with higher power.
- The two motors linked to the left-side driver are responsible for moving the base of the robotic arm.
- Meanwhile, the two motors on the right-side driver control the arm's movements, allowing it to pick and place objects effectively.



A circuit is essentially any loop through which matter flows. In the case of an electronic circuit, the matter in question is the charge carried by electrons, which originate from the positive terminal of a voltage source. When this charge travels from the positive terminal, moves around the loop, and finally reaches the negative terminal, we say the circuit is complete. However, this circuit is made up of various components that can influence the flow of charge in different ways. Some components may resist the flow, while others simply store or dissipate charge. Additionally, some components need an external energy source, while others provide energy themselves.

Result

The robotic arm did a great job responding to commands from the Bluetooth module, successfully carrying out basic pick and place tasks. The system operated smoothly with the help of an 8051 microcontroller (and also Arduino) along with motor drivers.